

A Dirac-Type Field Model for Primes

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Abstract

We propose a Dirac-type field model on the number line in which the non-trivial zeros of the Riemann zeta function act as spectral modes, prime numbers as negative charges, and prime gaps as positive artifacts. A smooth-cut construction $W(p)$ with a mirror map defines local particle–antiparticle pairs. Numerical experiments based on the first 1000 Riemann zeros show (i) GUE statistics of the spectrum, (ii) phase antisymmetry of mirror pairs, and (iii) resonance of large prime gaps with spectral modes, suggesting a spectral–geometric duality between zeros and primes.

1 Introduction

The connection between prime numbers and the zeros of the Riemann zeta function has motivated operator-theoretic interpretations since Hilbert–Pólya. In this work we develop a Dirac-type field model in which primes appear as discrete charges interacting with a spectral field defined by the zeta zeros.

2 Smooth Cuts and Mirror Primes

Let $H_3 = \{2^a 3^b\}$ and $H_5 = \{2^a 3^b 5^c\}$. For each prime p define

$$W(p) = (\max(\ell_3(p), \ell_5(p)), \min(u_3(p), u_5(p))), \quad (1)$$

where $\ell_s(p)$ and $u_s(p)$ denote the neighboring s -smooth numbers. Normalized coordinates θ_s yield a mirror map $\theta_s^*(p) = 1 - \theta_s(p)$ and a mirror prime $p^{\mathbb{G}}$ chosen by least-square matching in the (θ_3, θ_5) plane.

3 Dirac-Type Field

Define the field

$$\Phi(x) = \sum_{n \leq N} \Re \left(\frac{x^{1/2 + i\gamma_n}}{1/2 + i\gamma_n} \right), \quad (2)$$

where $\rho_n = 1/2 + i\gamma_n$ are the non-trivial zeros of $\zeta(s)$. The associated phase field is

$$S(x) = \sum_{n \leq N} e^{i\gamma_n \log x}, \quad \varphi(x) = \arg S(x). \quad (3)$$

4 Numerical Experiments

Using the first 1000 Riemann zeros we perform three tests:

1. **Spectral statistics:** Nearest-neighbor spacings of the zeros follow the GUE distribution (Wigner surmise).
2. **Mirror symmetry:** For mirror pairs $(p, p^{\mathbb{L}})$ we observe approximate phase antisymmetry $\varphi(p) \approx -\varphi(p^{\mathbb{L}})$ modulo 2π .
3. **Resonance of gaps:** Large prime gaps tend to occur when $\log m_k$ (midpoints of gaps) is close to a spectral period $2\pi/\gamma_n$.

5 Discussion

These findings support the interpretation of primes and gaps as charges and artifacts in a quantized number-theoretic field whose stability axis is the critical line $\Re(s) = 1/2$.

6 Outlook

Future work includes extension to larger datasets and reconstruction of the underlying operator from its spectral data.

References

- [1] A. M. Odlyzko, *The 10^{20} th zero of the Riemann zeta function and 40 million of its neighbors*, preprint.
- [2] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Symp. Pure Math. **24** (1973).